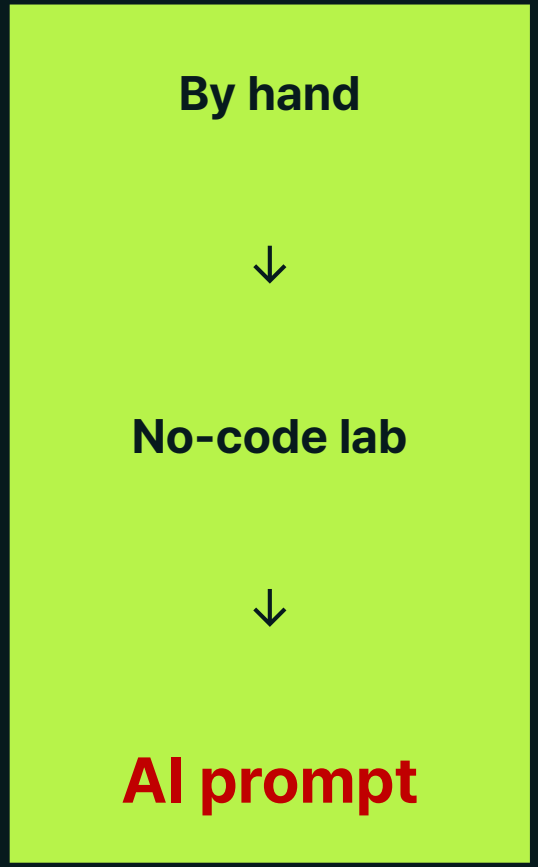
TFT · GNN · Agents Workbook

Step-by-step solutions you can follow with a calculator



Financial ML · Week 15 Workbook

Weights 1–6 give a forecast of about 11.69

Source values 10, 10.5, 9.8, 12, 11.2, 13.5

Step 1 · More weight on recent

$$\frac{1(10)+2(10.5)+3(9.8)+4(12)+5(11.2)+6(13.5)}{1+2+\cdots+6}=\frac{245.4}{21}\approx 11.69$$

Step 2 · Three methods

$$\frac{67}{6}\approx 11.17, \qquad \frac{245.4}{21}\approx 11.69, \qquad \text{last value } 13.5$$

Hand-set weights, so not a trained TFT result

A's signal reaches D after two synchronous updates

A=5, B=C=D=0 — links A–B, A–C, B–C, C–D

Step 1 · Source rule

$$h_v^{new} = 0.5\, h_v^{old} + 0.5 \times \text{mean}_{u \in N(v)}\, h_u^{old}$$

Step 2 · Round 1 (D = 0)

$$A_1 = 0.5(5) + 0.5 \cdot \tfrac{0+0}{2} = 2.5, \qquad B_1 = 0.5(0) + 0.5 \cdot \tfrac{5+0}{2} = 1.25, \qquad C_1 = 0.5 \cdot \tfrac{5+0+0}{3} = 0.833$$

Step 3 · 2 updates

$$A_2 = 0.5(2.5) + 0.5 \cdot \tfrac{1.25+0.833}{2} = 1.771, \qquad D_2 = 0.5(0) + 0.5(0.833) = 0.417$$

All nodes update at once from last-round values

One tool call lifts expected value from −2 to 7.5

Right +10, wrong −20, tool call −1 — accuracies illustrative

Step 1 · Memory vs. one search

Memory only: $0.6(10) - 0.4(20) = -2$

One retrieval: $0.95(10) - 0.05(20) - 1 = 7.5$

TFT, GNN and agents as input, process, output

Calculator and AI chat

1. Where

Use a calculator and AI chat.

2. Input

Enter 6 past values and GNN inputs.

3. Output

You get 3 forecasts and 2 node rounds.

4. Check

Check only last-round values are used.

Prompt for the AI chat

Explain TFT, GNN and AI agents to a non-major, with a table of each one's input, processing and output. State that the weighted average is a TFT intuition drill and the fixed neighbor average a message-passing drill. Compute the weighted average of 6 values and 2 synchronous updates on graph A-B-C-D. Finally, outline an earnings research flow that checks sources and units.

CALCULATE BY HAND.

CHECK THE DENOMINATOR.

THEN TRUST THE NUMBER.

Calculate once by hand
and the assumptions show.